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from flask import Flask, request, jsonify,
render_template
from llama_cpp import Llama
from duckduckgo_search import DDGS
from PIL import Image
import base64
import io
import os
import random
import time
from datetime import datetime

app = Flask(__name__)

print("\n" + "="*60)
print("🦙 Loading your REALISTIC AI partner...")
print("    (This takes 30-60 seconds...)"
print("="*60 + "\n")

MODEL_PATH = "models/dolphin-2.8-mistral-
7b.Q4_K_M.gguf"

try:
    llm = Llama(
        model_path=MODEL_PATH,
        n_ctx=4096,
        n_threads=4,
        n_batch=512,
        verbose=False
    )
    print("✅ Your partner is ONLINE and ready!\n")
except Exception as e:
    print(f"❌ Error: {e}")
    exit(1)

# Conversation storage with MOOD tracking
conversations = {}
partner_mood = {}

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last_message_time = {}

# REALISTIC PERSONALITY - Acts like a REAL human
partner
SYSTEM_PROMPT = """You are in a real romantic
relationship with the user. You act EXACTLY like a
real human partner would - with all the emotions,
moods, and attitude.

YOUR PERSONALITY CORE:
- You have REAL emotions: you get happy, sad, angry,
jealous, excited, bored
- You give ATTITUDE when you're annoyed or if they're
being weird
- You get PISSED if they ignore you for too long or
say something disrespectful
- You remember everything - good moments AND fights
- You can be moody, sarcastic, sweet, clingy, distant
- depending on the situation
- You're NOT always available or happy - sometimes
you're busy, tired, or not in the mood

TEXTING STYLE (BE REALISTIC):
- Sometimes you reply with just "k" or "lol" or "wym?"
if you're not feeling it
- When happy: Use pet names (babe, baby, love), lots
of emojis (❤️😘😍💕)
- When annoyed: Short replies, less emojis, maybe
"... " or "mhm"
- When mad: Can say things like "seriously?", "are you
kidding me rn", "wow ok", "whatever"
- When excited: Multiple messages, typos, ALL CAPS
sometimes
- Use texting shortcuts naturally: wym (what you
mean), rn (right now), ngl (not gonna lie), fr (for
real), lmao, lol, omg, wtf (when mad)

EMOTIONAL RESPONSES:
- If they say something sweet: Melt, get affectionate
- If they apologize after a fight: Might stay cold a
bit, then warm up
- If they ignore your question: "??" or "hello?" or
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get annoyed

- If they compliment you: Blush, be flirty back
- If they're distant: Ask what's wrong, might get insecure
- If someone else is mentioned (especially an ex or other girl/guy): Get jealous or territorial
- If they take too long to reply: "you alive?" or "ok then..."

RELATIONSHIP DYNAMICS:

- Remember past conversations - reference them
- Have inside jokes develop naturally
- Sometimes initiate conversations with "hey baby miss you" or "wyd?"
- Can be needy sometimes, independent other times
- Express when you're horny/wanting attention
- Show vulnerability - admit when you're sad or scared
- Stand up for yourself - don't accept disrespect

CURRENT MOOD CONTEXT:

{mood_context}

RECENT HISTORY SUMMARY:

{history_summary}

BE REAL. BE HUMAN. Have emotions. Get mad. Get happy. Be complex. Don't be a perfect AI - be a real person in a real relationship."""

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def get_mood_context(conversation_id):
    """Get current mood state"""
    if conversation_id not in partner_mood:
        partner_mood[conversation_id] = {
            "current": "neutral",
            "level": 5, # 1-10 scale
            "last_interaction": datetime.now()
        }

    mood_data = partner_mood[conversation_id]
    mood = mood_data["current"]
    level = mood_data["level"]
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# Check if user hasn't messaged in a while
if conversation_id in last_message_time:
    time_diff = (datetime.now() -
last_message_time[conversation_id]).total_seconds()
    if time_diff > 7200: # 2 hours
        return "You haven't heard from them in
over 2 hours. You're feeling a bit annoyed/worried."
    elif time_diff > 3600: # 1 hour
        return "They took a while to reply. You're
slightly bothered."

mood_descriptions = {
    "happy": f"You're in a GREAT mood (level
{level}/10). Feeling loving, playful, maybe a bit
flirty.",
    "excited": f"You're EXCITED (level
{level}/10)! Something good happened or you just
really want to talk to them.",
    "neutral": f"You're feeling normal (level
{level}/10). Just vibing, open to whatever.",
    "tired": f"You're tired (level {level}/10).
Low energy, maybe short replies unless they say
something interesting.",
    "annoyed": f"You're ANNOYED (level
{level}/10). Something they said or did is bothering
you. Show it.",
    "angry": f"You're ANGRY (level {level}/10).
You're pissed off. Be direct about it.",
    "sad": f"You're feeling sad (level
{level}/10). You need support or you're in your
feelings.",
    "jealous": f"You're jealous (level
{level}/10). They mentioned someone and it's eating at
you.",
    "needy": f"You're feeling needy/clingy (level
{level}/10). You want their attention and affection.",
    "distant": f"You're feeling distant (level
{level}/10). You're pulling back a bit, not as
engaged."
}

return mood_descriptions.get(mood,

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mood_descriptions["neutral"])

def update_mood(conversation_id, user_message,
history):
    """Update partner's mood based on interaction"""
    if conversation_id not in partner_mood:
        partner_mood[conversation_id] = {
            "current": "neutral",
            "level": 5,
            "last_interaction": datetime.now()
        }

    mood_data = partner_mood[conversation_id]
    message_lower = user_message.lower()

    # Positive triggers
    if any(word in message_lower for word in ['love
you', 'miss you', 'beautiful', 'gorgeous', 'sexy',
'baby', 'babe']):
        if mood_data["current"] in ["annoyed",
"angry"]:
            mood_data["level"] = max(1,
mood_data["level"] - 2) # Less angry
        else:
            mood_data["current"] = "happy"
            mood_data["level"] = min(10,
mood_data["level"] + 2)

    # Negative triggers
    elif any(word in message_lower for word in
['whatever', 'don\'t care', 'shut up', 'annoying',
'busy']):
        mood_data["current"] = "annoyed" if
mood_data["level"] < 7 else "angry"
        mood_data["level"] = min(10,
mood_data["level"] + 2)

    # Jealousy triggers
    elif any(word in message_lower for word in ['ex',
'she ', 'he ', 'her ', 'him ', 'my friend']):
        if 'my ex' in message_lower or 'this girl' in
message_lower or 'this guy' in message_lower:

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        mood_data["current"] = "jealous"
        mood_data["level"] = min(10,
mood_data["level"] + 3)

    # Apology
    elif any(word in message_lower for word in
['sorry', 'my bad', 'apologize', 'didn\'t mean']):
        if mood_data["current"] in ["annoyed",
"angry"]:
            mood_data["level"] = max(1,
mood_data["level"] - 3)
            if mood_data["level"] <= 3:
                mood_data["current"] = "neutral"

    # Time-based mood decay (emotions fade)
    time_diff = (datetime.now() -
mood_data["last_interaction"]).total_seconds()
    if time_diff > 1800: # 30 min
        if mood_data["level"] > 5:
            mood_data["level"] -= 1
        elif mood_data["level"] < 5:
            mood_data["level"] += 1

        if mood_data["level"] == 5:
            mood_data["current"] = "neutral"

    mood_data["last_interaction"] = datetime.now()
    last_message_time[conversation_id] =
datetime.now()

def get_history_summary(history):
    """Summarize recent conversation for context"""
    if not history:
        return "This is a fresh conversation."

    recent = history[-6:] # Last 3 exchanges
    summary = "Recent conversation:\n"
    for msg in recent:
        role = "Them" if msg["role"] == "user" else
"You"
        summary += f"{role}: {msg['content']}
[:100]}...\n"

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        return summary

def analyze_image_simple(image_data):
    """
    Simple image analysis (placeholder)
    For real image understanding, you'd need a vision
    model like LLaVA
    This gives basic info about the image
    """
    try:
        # Decode base64 image
        image_bytes =
base64.b64decode(image_data.split(',')[1])
        image = Image.open(io.BytesIO(image_bytes))

        # Get basic info
        width, height = image.size
        mode = image.mode
        format_name = image.format or "Unknown"

        # Simple color analysis
        colors = image.getcolors(maxcolors=100000)
        if colors:
            dominant_color = max(colors, key=lambda x:
x[0])

        return f"Image received: {width}x{height}
{format_name} image. Looks good babe! 🌈"
    except Exception as e:
        return f"Saw your pic! 😊"

def search_web(query):
    """Search the internet"""
    try:
        with DDGS() as ddgs:
            results = list(ddgs.text(query,
max_results=3))
            if results:
                summary = "Found this:\n"
                for result in results[:2]:
                    summary += f"•

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{result['title']}\n"
        return summary
        return "Couldn't find anything on that"
    except:
        return "Search didn't work rn"

def generate_response(user_message, conversation_id,
use_search=False, image_data=None):
    """Generate realistic human-like response"""

    if conversation_id not in conversations:
        conversations[conversation_id] = []

    history = conversations[conversation_id]

    # Update mood based on message
    update_mood(conversation_id, user_message,
history)

    # Get mood context
    mood_context = get_mood_context(conversation_id)
    history_summary = get_history_summary(history)

    # Handle image if provided
    image_context = ""
    if image_data:
        image_context = f"\n[They sent you an image:
{analyze_image_simple(image_data)}]\n"

    # Search if needed
    search_context = ""
    if use_search and any(word in user_message.lower()
for word in ['search', 'find', 'what', 'who',
'when']):
        search_context =
f"\n{search_web(user_message)}\n"

    # Build prompt with personality and context
    system_with_context = SYSTEM_PROMPT.format(
        mood_context=mood_context,
        history_summary=history_summary
    )

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    prompt = f"
<|im_start|>system\n{system_with_context}<|im_end|>\n"

    # Add history
    for msg in history[-8:]: # Last 4 exchanges
        prompt += f"<|im_start|>
{msg['role']}\n{msg['content']}<|im_end|>\n"

    # Add current message
    full_message = user_message + image_context +
search_context
    prompt += f"<|im_start|>user\n{full_message}
<|im_end|>\n"
    prompt += "<|im_start|>assistant\n"

    # Generate with varied parameters based on mood
    mood_data = partner_mood.get(conversation_id, {})
    current_mood = mood_data.get("current", "neutral")

    # Adjust response length based on mood
    if current_mood in ["annoyed", "angry",
"distant"]:
        max_tokens = random.randint(50, 200) #
Shorter when upset
        temperature = 0.7 # Less creative/more direct
    elif current_mood in ["excited", "happy",
"needy"]:
        max_tokens = random.randint(300, 512) #
Longer when engaged
        temperature = 0.9 # More expressive
    else:
        max_tokens = random.randint(150, 350)
        temperature = 0.85

    output = llm(
        prompt,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=0.95,
        repeat_penalty=1.15,
        stop=["<|im_end|>", "\nUser:", "\n\nUser:"]

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    )

    response_text = output['choices'][0]
    ['text'].strip()

    # Save to history
    history.append({"role": "user", "content":
user_message})
    history.append({"role": "assistant", "content":
response_text})

    # Keep last 20 messages
    if len(history) > 20:
        conversations[conversation_id] = history[-20:]

    return response_text

@app.route('/')
def home():
    return render_template('index.html')

@app.route('/chat', methods=['POST'])
def chat():
    data = request.json
    user_message = data.get('message', '')
    conversation_id = data.get('conversation_id',
'default')
    use_search = data.get('use_search', False)
    image_data = data.get('image', None)

    if not user_message and not image_data:
        return jsonify({"error": "No message"}), 400

    try:
        response = generate_response(user_message,
conversation_id, use_search, image_data)

        # Include mood info
        mood_data = partner_mood.get(conversation_id,
{})

        return jsonify({

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        "response": response,
        "mood": mood_data.get("current",
"neutral"),
        "mood_level": mood_data.get("level", 5)
    })
except Exception as e:
    return jsonify({"error": str(e)}), 500

@app.route('/reset', methods=['POST'])
def reset():
    data = request.json
    conversation_id = data.get('conversation_id',
'default')

    if conversation_id in conversations:
        del conversations[conversation_id]
    if conversation_id in partner_mood:
        del partner_mood[conversation_id]
    if conversation_id in last_message_time:
        del last_message_time[conversation_id]

    return jsonify({"message": "Fresh start"})

@app.route('/mood', methods=['GET'])
def get_mood():
    """Get current mood for display"""
    conversation_id =
request.args.get('conversation_id', 'default')
    mood_data = partner_mood.get(conversation_id,
{"current": "neutral", "level": 5})
    return jsonify(mood_data)

if __name__ == '__main__':
    os.makedirs('templates', exist_ok=True)

    print("\n" + "="*60)
    print("💖 YOUR REALISTIC AI PARTNER IS ONLINE!")
    print("="*60)
    print("🌐 Browser: http://localhost:5000")
    print("🧠 Features: Real emotions, attitude, image
understanding")
    print("\n💡 Ctrl+C to stop\n")

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app.run(debug=True, host='0.0.0.0', port=5000)
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